

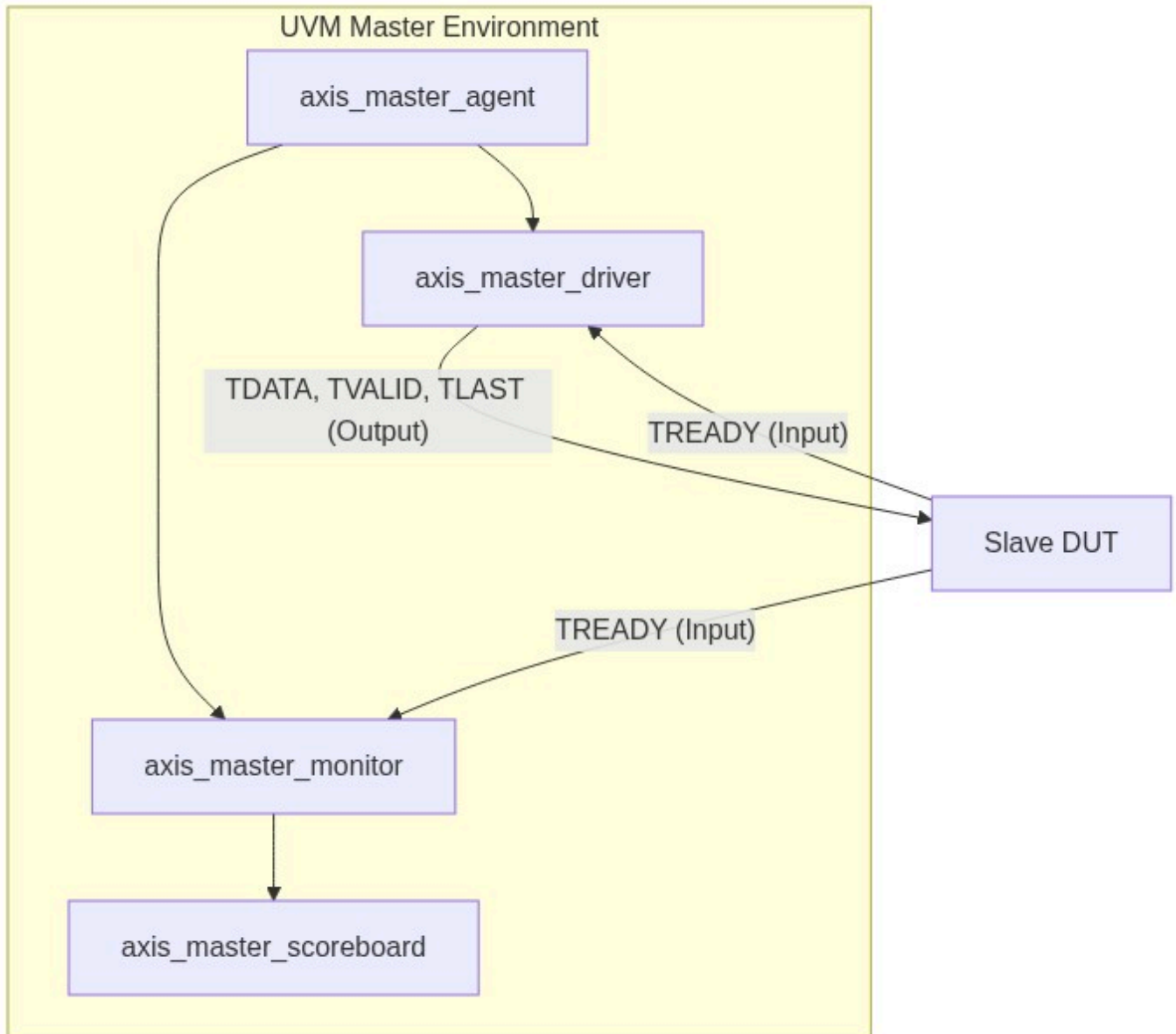
Verification Plan Report

Project: AXI-Stream Role-Safe Verification

Version: 3.1

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MS_01	Handshake (Stability)	Master Data Stability	While TVALID is HIGH and TREADY (from DUT) is observed to be LOW, the Master VIP MUST keep TDATA and all sidebands stable.

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REQ_MS_02	Handshake (Independence)	TVALID/TREADY Independence	Master VIP MUST NOT wait for TREADY assertion before asserting TVALID.
REQ_MS_03	Reset (Suppression)	Reset TVALID LOW	Master VIP MUST drive TVALID LOW during reset. TREADY (Input) is ignored per protocol.
REQ_MS_04	Packet (TLAST)	End-of-Packet Marker	Master VIP MUST drive TLAST HIGH to indicate the final beat of a packet.
REQ_MS_05	Packet (Consistency)	ID/DEST Integrity	Master VIP MUST hold TID and TDEST constant while TLAST is LOW in an active transfer.
REQ_MS_06	Monitoring (Slave Compliance)	Slave Ready Monitoring	Master VIP Monitor SHALL verify that Slave DUT drives TREADY to a valid logic level (no X-states).

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MS_01	Valid-Wait-Ready: Master VIP asserts TVALID=1. Slave DUT is observed holding TREADY=0. Verify Master VIP keeps TDATA stable for 10 cycles.	REQ_MS_01
TC_MS_02	Instant Handshake: Master VIP asserts TVALID=1. Slave DUT is observed already at TREADY=1. Verify transfer completes in a single cycle.	REQ_MS_02
TC_MS_03	Back-Pressure Stress: Master VIP drives a 100-beat burst. Slave DUT is observed to toggle TREADY randomly. Verify zero data loss.	REQ_MS_01, REQ_MS_06
TC_MS_04	Reset Suppression: Verify Master VIP drives TVALID=0 while ARESETn is LOW, even if Slave DUT drives TREADY=1.	REQ_MS_03
TC_MS_05	TID Stability Check: Master VIP drives a packet. Monitor observes TREADY stall cycle 3. Verify Master VIP TID remains frozen.	REQ_MS_05
TC_MS_06	Single-Beat Packet: Master VIP drives TLAST=1 and TVALID=1 simultaneously. Waits for Slave DUT TREADY.	REQ_MS_04
TC_MS_07	ZBT Packet: Master VIP drives TLAST=1 with all TKEEP=0. Observe Slave DUT handshake.	REQ_MS_04
TC_MS_08	Slave Readiness Observation: Sequence drives multi-beat. Monitor verifies Slave DUT eventually asserts TREADY (Liveness check).	REQ_MS_06
TC_MS_09	Slave Stall Analysis: Master VIP drives 10 packets. Monitor logs performance metric based on Slave DUT's TREADY latency.	REQ_MS_06

4. Functional Coverage Groups

Group Name	Description
cg_master_handshake	Cross of Master TVALID x Observed Slave TREADY.

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cg_packet_boundary	TLAST transitions during Master-led burst.
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5. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Output	TD_W	Data payload from Master.
TVALID	Output	1	Valid from Master.
TREADY	Input	1	Ready from Slave DUT (Observed by Master).
TLAST	Output	1	End of packet from Master.
ACLK	Input	1	Global clock.
ARESETn	Input	1	Global reset.

6. Verification Environment Structure

```
verif/axis_role_safe_v3_1/
```

```
|-- agent/
|   |-- axis_master_driver.sv
|   |-- axis_master_monitor.sv
|   |-- axis_master_agent.sv
|   |-- axis_if.sv
|-- env/
    |-- axis_env.sv
    |-- axis_scoreboard.sv
```

7. SystemVerilog Implementation

Master Driver: Handshake Protocol

```
task axis_master_driver::drive_transfer(axis_tx tr);
    vif.TVALID <= 1'b1;
    vif.TDATA  <= tr.data;
    vif.TLAST  <= tr.last;
    // Wait for Slave DUT to respond with TREADY
    do begin
        @(posedge vif.ACLK);
    end while (vif.TREADY != 1'b1);
    vif.TVALID <= 1'b0;
endtask
```